

Java Assignment: 1

1. What is Java?

Java is a high-level, object-oriented programming language used to build applications.

2. Is Java platform dependent or independent?

Java is platform independent.

3. What is JVM?

JVM (Java Virtual Machine) runs Java bytecode on any system.

4. What is JDK and JRE?

- JDK: Development kit (compiler + tools)
- JRE: Runtime environment (JVM + libraries)

5. Why is Java called “Write Once, Run Anywhere”?

Because Java code runs on any system with JVM.

6. What is a class in Java?

A blueprint to create objects.

7. What is an object?

An instance of a class.

8. What is the main method in Java?

Entry point of program:

```
public static void main(String[] args)
```

9. What is the use of System.out.println()?

Prints output to console.

11. What are data types in Java?

They define the type of data a variable can store.

12. Examples of primitive data types

int, float, double, char, boolean, byte, short, long

13. What is a variable?

A container to store data.

14. How to declare a variable?

```
int x = 10;
```

15. Difference between int and float?

- int → whole numbers
- float → decimal numbers

16. What is type casting?

Converting one data type to another.

17. Default value of int?

0

18. What is a keyword in Java?

Reserved word with special meaning (e.g., class, if)

19. Can we use keywords as variable names?

No

20. Difference between = and ==?

- = → assignment
- == → comparison

21. What is an if statement?

Used for decision making.

22. Difference between if and switch?

- if → multiple conditions
- switch → multiple fixed values

23. What is a loop?

Used to repeat code.

24. Types of loops in Java?

For, while, do-while

25. Difference between for and while loop?

- for → known iterations
- while → condition-based

26. What is break statement?

Stops loop immediately.

27. What is continue statement?

Skips current iteration.

28. What if main method is missing?

Program won't run.

29. Can we run Java without class?

No (class is required).

30. What if we don't compile Java program?

It won't run (needs bytecode).

31. Can a variable change its data type?

No (Java is strongly typed).

32. Size of int in Java?

4 bytes (32 bits)

33. Why Java does not support pointers directly?

For security and simplicity.